

Identifying adaptive alleles to assess genetic rescue potential in an alpine plant

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1 Introduction

Climate change is rapidly altering the abiotic and biotic environments of natural plant populations, threatening their persistence. While populations may respond through migration or adaptation, alpine plants are often constrained by topography and limited dispersal capacity. Consequently, gene flow from warm-adapted low-elevation populations may play a key role in facilitating adaptation to climate change in high-elevation populations.

This process, known as genetic rescue, could enhance the persistence of alpine plants under future climatic conditions. In this project, we investigate whether gene flow from low-elevation ecotypes can buffer high-elevation populations of *Dianthus carthusianorum* against climate-driven local extinction.

Using a field experiment along an elevational gradient in the Swiss Alps, we simulate future warming scenarios and novel competitive interactions expected under climate change.

This first part of the project focuses on identifying adaptive alleles differentiated between low- and high-elevation populations, characterizing allele frequency shifts along elevation gradients, and assessing allelic turnover in relation to climatic and biotic factors.



Dianthus carthusianorum

2 Materials & Methods

Seventeen populations of *Dianthus carthusianorum* were sampled along four elevational gradients in the central Alps. Whole-genome resequencing will be performed for 20 individuals per population.

Population genetic structure will first be characterized using principal component analysis (PCA) and ADMIXTURE based on genome-wide SNP data.

Candidate adaptive loci will then be identified using:

- Genome scans incorporating demographic models, and
- Genotype-environment association analyses using redundancy analysis (RDA).

Finally, allelic turnover along climatic and elevational gradients will be assessed using:

- Generalized dissimilarity modelling (GDM), and
- GradientForest analyses.

3 Expected Results

We expect to observe genomic differentiation between high and low elevation ecotypes at adaptive loci and admixture among populations along elevational gradients.

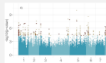


Example of PCA on genetic data along an environmental cline (from Van Deurs et al. 2025)



Example of admixture along an environmental cline (from Homolová et al. 2024)

Genome scans and genotype-environment association analyses should identify candidate adaptive loci associated with climatic variables, particularly temperature-related gradients.



Example of candidate adaptive allele identification via genotype-environment association analysis (from Van Deurs et al. 2025)

We further expect allele frequencies at candidate adaptive loci to shift predictably across elevation, revealing non-random allelic turnover linked to climatic and potentially biotic factors.



Example of allelic turnover of candidate adaptive SNPs along an environmental gradient (from Van Deurs et al. 2025)

4 Conclusion

Identifying climate-associated adaptive loci and allelic turnover across elevational gradients will help determine whether warm-adapted alleles from low-elevation populations can facilitate adaptive gene flow and contribute to the genetic rescue of high-elevation populations of *Dianthus carthusianorum* under climate change.

References

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2. Homolová, M., Lenoir, M.C., Marková, S. et al. Genetic admixture drives climate adaptation in the bank voles. *Genome Biol.* 7, 953 (2024). <https://doi.org/10.1038/s41588-024-0049-6>